notebook_1

Notebook 1: Data Understanding and Type Validation

This notebook verifies that the processed soccer data loads correctly, has the expected column classes, contains plausible example values, and obeys the basic match-outcome logic needed for modeling.

The goal is not to build features or train a model yet. The goal is to understand and validate the data.

1. Processed files

```r
processed_files <- c(
  statsbomb_competitions = file.path(PROCESSED_DIR, "statsbomb_competitions.csv"),
  statsbomb_matches = file.path(PROCESSED_DIR, "statsbomb_matches.csv"),
  football_data_uk_matches = file.path(PROCESSED_DIR, "football_data_uk_matches.csv"),
  international_results = file.path(PROCESSED_DIR, "international_results.csv")
)

file_check <- tibble::tibble(
  dataset = names(processed_files),
  path = unname(processed_files),
  exists = file.exists(processed_files),
  file_size_mb = round(file.info(processed_files)$size / 1024^2, 3)
)

knitr::kable(
  file_check,
  caption = "Processed data files expected by this notebook"
)
```

Table 1: Processed data files expected by this notebook

dataset	path	exists	file_size_mb
statsbomb_competitions	/Users/marco/Documents/Projects/worldcup-forecast-r/data/processed/statsbomb_competitions.csv	TRUE	0.010
statsbomb_matches	/Users/marco/Documents/Projects/worldcup-forecast-r/data/processed/statsbomb_matches.csv	TRUE	0.948
football_data_uk_matches	/Users/marco/Documents/Projects/worldcup-forecast-r/data/processed/football_data_uk_matches.csv	TRUE	38.677
international_results	/Users/marco/Documents/Projects/worldcup-forecast-r/data/processed/international_results.csv	TRUE	13.029

```r
if (any(!file_check$exists)) {
  stop(
    "One or more processed files are missing. Run the full pipeline before knitting this notebook."
  )
}
```

2. Load processed data using project column specifications

This section loads each processed CSV using `processed_csv_col_types()` from `src/02_helpers.R`.

```r
processed_data <- purrr::imap(
  processed_files,
  ~ readr::read_csv(
    file = .x,
    col_types = processed_csv_col_types(.x),
    show_col_types = FALSE
  )
)

names(processed_data)
```

```
## [1] "statsbomb_competitions"   "statsbomb_matches"
## [3] "football_data_uk_matches" "international_results"
```

3. Dataset inventory

```r
dataset_inventory <- purrr::imap_dfr(
  processed_data,
  function(dat, dataset_name) {
    tibble::tibble(
      dataset = dataset_name,
      rows = nrow(dat),
      columns = ncol(dat),
      first_date = if ("date" %in% names(dat)) as.character(min(dat$date, na.rm = TRUE)) else NA_charac
      last_date = if ("date" %in% names(dat)) as.character(max(dat$date, na.rm = TRUE)) else NA_charact
    )
  }
)

knitr::kable(
  dataset_inventory,
  caption = "Dataset inventory"
)
```

Table 2: Dataset inventory

dataset	rows	columns	first_date	last_date
statsbomb_competitions	80	12	NA	NA
statsbomb_matches	3961	26	1958-06-24	2025-07-27
football_data_uk_matches	177295	36	1993-07-23	2024-06-02
international_results	49257	21	1872-11-30	2026-03-31

4. All columns, classes, and examples

This table shows every column in every processed dataset, its R class, its storage type, missing count, unique count, and a few example values.

```r
class_summary <- purrr::imap_dfr(
  processed_data,
  function(dat, dataset_name) {
    tibble::tibble(
      dataset = dataset_name,
      column_position = seq_along(names(dat)),
      column = names(dat),
      class = purrr::map_chr(dat, ~ paste(class(.x), collapse = " / ")),
      typeof = purrr::map_chr(dat, typeof),
      n_missing = purrr::map_int(dat, ~ sum(is.na(.x))),
      n_unique = purrr::map_int(dat, dplyr::n_distinct),
      example_values = purrr::map_chr(
        dat,
        ~ paste(utils::head(unique(na.omit(.x)), 5), collapse = ", ")
      )
    )
  }
)

knitr::kable(
  class_summary,
  caption = "Observed classes and example values for all processed columns"
)
```

Table 3: Observed classes and example values for all processed columns

dataset	column_position	column	class	typeof	n_missing	n_unique	example_values
statsbomb_competitions	1	competition_id	integer	integer	0	24	9, 1267, 16, 223, 87
statsbomb_competitions	2	season_id	integer	integer	0	48	281, 27, 107, 4, 1
statsbomb_competitions	3	country_name	character	character	0	13	Germany, Africa, Europe, South America, Spain
statsbomb_competitions	4	competition_name	character	character	0	24	1. Bundesliga, African Cup of Nations, Champions League, Copa America, Copa del Rey
statsbomb_competitions	5	competition_gender	character	character	0	2	male, female
statsbomb_competitions	6	competition_youth	logical	logical	0	2	FALSE, TRUE
statsbomb_competitions	7	competition_international	logical	logical	0	2	FALSE, TRUE
statsbomb_competitions	8	season_name	character	character	0	48	2023/2024, 2015/2016, 2023, 2018/2019, 2017/2018

dataset	column_position	column	class	type	n_missing	n_unique	example_values
statsbomb_competitions	9	match_updated	character	character	0	80	2024-09-28T20:46:38.893391, 2024-05-19T11:11:14.192381, 2026-05-12T21:18:08.827431, 2026-05-15T15:54:04.598614, 2024-02-13T02:35:28.134882
statsbomb_competitions	10	match_updated_360	character	character	22	16	2025-11-15T23:17:41.827093, 2026-05-02T02:07:18.902396, 2021-06-13T16:17:31.694, 2026-05-04T01:53:40.309717, 2021-06-12T16:17:31.694
statsbomb_competitions	11	match_available_360	character	character	68	13	2025-11-15T23:17:41.827093, 2026-05-02T02:07:18.902396, 2026-05-04T01:53:40.309717, 2025-11-02T03:55:53.923193, 2026-04-06T18:59:05.942748
statsbomb_competitions	12	match_available	character	character	0	80	2024-09-28T20:46:38.893391, 2024-05-19T11:11:14.192381, 2026-05-12T21:18:08.827431, 2026-05-15T15:54:04.598614, 2024-02-13T02:35:28.134882
statsbomb_matches	1	source	character	character	0	1	StatsBomb Open Data
statsbomb_matches	2	raw_file	character	character	0	80	/Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/116_68.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_1.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_2.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_21.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_22.json
statsbomb_matches	3	source_match_id	character	character	0	3961	3750234, 9880, 9912, 9924, 9855
statsbomb_matches	4	date	Date	double	0	1362	1977-08-28, 2018-04-14, 2018-04-29, 2018-05-06, 2018-03-18
statsbomb_matches	5	season	character	character	0	48	1977, 2017/2018, 2016/2017, 2009/2010, 2010/2011
statsbomb_matches	6	competition	character	character	0	24	North American League, La Liga, Indian Super league, African Cup of Nations, Serie A
statsbomb_matches	7	home_team	character	character	0	344	NY Cosmos, Barcelona, RC Deportivo La Coruña, Las Palmas, Eibar
statsbomb_matches	8	away_team	character	character	0	343	Seattle Sounders, Valencia, Barcelona, Real Madrid, Athletic Club
statsbomb_matches	9	home_score	integer	integer	0	13	2, 1, 0, 3, 6
statsbomb_matches	10	away_score	integer	integer	0	10	1, 4, 2, 0, 3
statsbomb_matches	11	competition_id	integer	integer	0	24	116, 11, 1238, 1267, 12
statsbomb_matches	12	season_id	integer	integer	0	48	68, 1, 2, 21, 22
statsbomb_matches	13	match_week	integer	integer	0	38	1, 32, 35, 36, 29
statsbomb_matches	14	competition_country	character	character	0	13	North and Central America, Spain, India, Africa, Italy
statsbomb_matches	15	stadium	character	character	10	328	Providence Park, Spotify Camp Nou, Estadio Abanca-Riazor, Estadio de Gran Canaria, Estadio Municipal de Ipurúa
statsbomb_matches	16	data_version	character	character	2	4	1.1.0, 1.0.3, 1.0.2
statsbomb_matches	17	shot_fidelity_version	character	character	1975	2	2
statsbomb_matches	18	xy_fidelity_version	character	character	2463	2	2
statsbomb_matches	19	match_result	character	character	0	3	H, A, D
statsbomb_matches	20	result_class	integer	integer	0	3	1, -1, 0
statsbomb_matches	21	home_win	integer	integer	0	2	1, 0
statsbomb_matches	22	draw	integer	integer	0	2	0, 1
statsbomb_matches	23	away_win	integer	integer	0	2	0, 1
statsbomb_matches	24	goal_difference	integer	integer	0	21	1, -2, 0, 2, 3
statsbomb_matches	25	total_goals	integer	integer	0	14	3, 6, 4, 2, 7
statsbomb_matches	26	neutral	logical	logical	3961	1	
football_data_uk_matches	1	source	character	character	0	1	football-data.co.uk

dataset	column_position	column	class	type	n_missing	n_unique	sample_values
football_data_uk_matches	2	raw_file	character	character	0	457	/Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/D1.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/D2.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/E0.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/E1.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/E2.csv
football_data_uk_matches	3	source_match_id	character	character	0	177295	D1_0001_1, D1_0001_2, D1_0001_3, D1_0001_4, D1_0001_5
football_data_uk_matches	4	source_league_code	character	character	0	15	D1, D2, E0, E1, E2
football_data_uk_matches	5	source_season_code	character	character	0	31	0001, 0102, 0203, 0304, 0405
football_data_uk_matches	6	date	Date	double	0	7771	2000-08-11, 2000-08-12, 2000-08-13, 2000-08-18, 2000-08-19
football_data_uk_matches	7	season	character	character	0	31	0001, 0102, 0203, 0304, 0405
football_data_uk_matches	8	competition	character	character	0	15	D1, D2, E0, E1, E2
football_data_uk_matches	9	home_team	character	character	0	564	Dortmund, Bayern Munich, Freiburg, Hamburg, Kaiserslautern
football_data_uk_matches	10	away_team	character	character	0	564	Hansa Rostock, Hertha, Stuttgart, Munich 1860, Bochum
football_data_uk_matches	11	home_score	integer	integer	0	11	1, 4, 2, 0, 3
football_data_uk_matches	12	away_score	integer	integer	0	11	0, 1, 2, 4, 3
football_data_uk_matches	13	full_time_result	character	character	0	3	H, D, A
football_data_uk_matches	14	half_time_home_score	numeric	double	20040	10	0, 1, 2, 3, 4
football_data_uk_matches	15	half_time_away_score	numeric	double	20040	12	0, 2, 1, 3, 4
football_data_uk_matches	16	half_time_result	character	character	20041	4	D, H, A
football_data_uk_matches	17	home_shots	numeric	double	79199	47	17, 14, 15, 18, 11
football_data_uk_matches	18	away_shots	numeric	double	79196	41	5, 11, 18, 9, 3
football_data_uk_matches	19	home_shots_on_target	numeric	double	80112	27	7, 6, 5, 2, 3
football_data_uk_matches	20	away_shots_on_target	numeric	double	80112	23	2, 5, 7, 3, 9
football_data_uk_matches	21	home_corners	numeric	double	79572	27	7, 4, 5, 6, 3
football_data_uk_matches	22	away_corners	numeric	double	79572	24	3, 9, 7, 5, 4
football_data_uk_matches	23	home_fouls	numeric	double	80336	42	25, 13, 22, 0, 9
football_data_uk_matches	24	away_fouls	numeric	double	80336	42	19, 12, 17, 0, 8
football_data_uk_matches	25	home_yellow_cards	numeric	double	79015	13	1, 2, 0, 3, 4
football_data_uk_matches	26	away_yellow_cards	numeric	double	79014	12	5, 0, 1, 2, 3
football_data_uk_matches	27	home_red_cards	numeric	double	79015	4	0, 1, 2, 3
football_data_uk_matches	28	away_red_cards	numeric	double	79014	7	0, 1, 2, 3, 4
football_data_uk_matches	29	match_result	character	character	0	3	H, D, A
football_data_uk_matches	30	result_class	integer	integer	0	3	1, 0, -1
football_data_uk_matches	31	home_win	integer	integer	0	2	1, 0
football_data_uk_matches	32	draw	integer	integer	0	2	0, 1
football_data_uk_matches	33	away_win	integer	integer	0	2	0, 1
football_data_uk_matches	34	goal_difference	integer	integer	0	21	1, 3, 4, 0, -1
football_data_uk_matches	35	total_goals	integer	integer	0	14	1, 5, 4, 2, 3
football_data_uk_matches	36	neutral	logical	logical	177295	1	
international_results	1	source	character	character	0	1	international_results
international_results	2	raw_file	character	character	0	1	/Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/international_results/results.csv
international_results	3	source_match_id	character	character	0	49257	1872_11_30_scotland_england_friendly, 1873_03_08_england_scotland_friendly, 1874_03_07_scotland_england_friendly, 1875_03_06_england_scotland_friendly, 1876_03_04_scotland_england_friendly

dataset	column_position	column	class	type	n_missing	n_unique	sample_values
international_results	4	date	Date	double	0	16434	1872-11-30, 1873-03-08, 1874-03-07, 1875-03-06, 1876-03-04
international_results	5	season	character	character	0	155	1872, 1873, 1874, 1875, 1876
international_results	6	competition	character	character	0	198	Friendly, British Home Championship, Évence Coppée Trophy, Muratti Vase, Copa Lipton
international_results	7	home_team	character	character	0	327	Scotland, England, Wales, Northern Ireland, United States
international_results	8	away_team	character	character	0	321	England, Scotland, Wales, Northern Ireland, Canada
international_results	9	home_score	integer	integer	0	26	0, 4, 2, 3, 1
international_results	10	away_score	integer	integer	0	22	0, 2, 1, 3, 4
international_results	11	match_result	character	character	0	3	D, H, A
international_results	12	result_class	integer	integer	0	3	0, 1, -1
international_results	13	home_win	integer	integer	0	2	0, 1
international_results	14	draw	integer	integer	0	2	1, 0
international_results	15	away_win	integer	integer	0	2	0, 1
international_results	16	goal_difference	integer	integer	0	47	0, 2, 1, 3, 4
international_results	17	total_goals	integer	integer	0	26	0, 6, 3, 4, 2
international_results	18	neutral	logical	logical	0	2	FALSE, TRUE
international_results	19	tournament	character	character	0	198	Friendly, British Home Championship, Évence Coppée Trophy, Muratti Vase, Copa Lipton
international_results	20	city	character	character	0	2138	Glasgow, London, Wrexham, Blackburn, Belfast
international_results	21	country	character	character	0	269	Scotland, England, Wales, Ireland, United States

5. Expected versus actual classes

The purpose of this check is to verify whether the processed files were parsed into the intended R classes.

```r
expected_common_match_classes <- tibble::tribble(
  ~column, ~expected_class,
  "source", "character",
  "raw_file", "character",
  "source_match_id", "character",
  "date", "Date",
  "season", "character",
  "competition", "character",
  "home_team", "character",
  "away_team", "character",
  "home_score", "integer",
  "away_score", "integer",
  "match_result", "character",
  "result_class", "integer",
  "home_win", "integer",
  "draw", "integer",
  "away_win", "integer",
  "goal_difference", "integer",
  "total_goals", "integer",
  "neutral", "logical"
)

expected_classes <- dplyr::bind_rows(
  expected_common_match_classes |>
    dplyr::mutate(dataset = "statsbomb_matches"),
```

```r
  expected_common_match_classes |>
    dplyr::mutate(dataset = "football_data_uk_matches"),

  expected_common_match_classes |>
    dplyr::mutate(dataset = "international_results"),

  tibble::tribble(
    ~column, ~expected_class,
    "competition_id", "integer",
    "season_id", "integer",
    "country_name", "character",
    "competition_name", "character",
    "competition_gender", "character",
    "competition_youth", "logical",
    "competition_international", "logical",
    "season_name", "character",
    "match_updated", "character",
    "match_updated_360", "character",
    "match_available_360", "character",
    "match_available", "character"
  ) |>
    dplyr::mutate(dataset = "statsbomb_competitions")
)

expected_classes <- expected_classes |>
  dplyr::select(dataset, column, expected_class)
```

```r
class_validation <- class_summary |>
  dplyr::select(
    dataset,
    column,
    actual_class = class,
    typeof,
    n_missing,
    n_unique,
    example_values
  ) |>
  dplyr::left_join(
    expected_classes,
    by = c("dataset", "column")
  ) |>
  dplyr::mutate(
    expected_class = dplyr::coalesce(expected_class, "not specified"),
    class_ok = dplyr::case_when(
      expected_class == "not specified" ~ NA,
      actual_class == expected_class ~ TRUE,
      stringr::str_starts(actual_class, paste0(expected_class, " /")) ~ TRUE,
      TRUE ~ FALSE
    )
  ) |>
  dplyr::arrange(dataset, dplyr::desc(class_ok == FALSE), column)

knitr::kable(
```

```r
  class_validation,
  caption = "Expected versus actual column classes"
)
```

Table 4: Expected versus actual column classes

dataset	column	actual_class	typeof	n_missing	n_unique	example_values	expected_class	class_ok
football_data_matches	away_goals	integer	integer	0	11	0, 1, 2, 4, 3	integer	TRUE
football_data_matches	away_team	character	character	0	564	Hansa Rostock, Hertha, Stuttgart, Munich 1860, Bochum	character	TRUE
football_data_matches	away_win	integer	integer	0	2	0, 1	integer	TRUE
football_data_matches	competition	character	character	0	15	D1, D2, E0, E1, E2	character	TRUE
football_data_matches	date	Date	double	0	7771	2000-08-11, 2000-08-12, 2000-08-13, 2000-08-18, 2000-08-19	Date	TRUE
football_data_matches	draw	integer	integer	0	2	0, 1	integer	TRUE
football_data_matches	goal_difference	integer	integer	0	21	1, 3, 4, 0, -1	integer	TRUE
football_data_matches	home_goals	integer	integer	0	11	1, 4, 2, 0, 3	integer	TRUE
football_data_matches	home_team	character	character	0	564	Dortmund, Bayern Munich, Freiburg, Hamburg, Kaiserslautern	character	TRUE
football_data_matches	home_win	integer	integer	0	2	1, 0	integer	TRUE
football_data_matches	match_result	character	character	0	3	H, D, A	character	TRUE
football_data_matches	neutral	logical	logical	177295	[illegible]		logical	TRUE
football_data_matches	raw_file	character	character	0	457	/Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/D1.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/D2.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/E0.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/E1.csv, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/football_data_uk/0001/E2.csv	character	TRUE
football_data_matches	result_class	integer	integer	0	3	1, 0, -1	integer	TRUE
football_data_matches	season	character	character	0	31	0001, 0102, 0203, 0304, 0405	character	TRUE
football_data_matches	source	character	character	0	1	football-data.co.uk	character	TRUE
football_data_matches	source_match_id	character	character	0	177295	D1_0001_1, D1_0001_2, D1_0001_3, D1_0001_4, D1_0001_5	character	TRUE
football_data_matches	total_goals	integer	integer	0	14	1, 5, 4, 2, 3	integer	TRUE
football_data_matches	away_corners	numeric	double	79572	24	3, 9, 7, 5, 4	not specified	NA
football_data_matches	away_fouls	numeric	double	80336	42	19, 12, 17, 0, 8	not specified	NA
football_data_matches	away_red_cards	numeric	double	79017	4	0, 1, 2, 3, 4	not specified	NA
football_data_matches	away_shots	numeric	double	79196	61	5, 11, 18, 9, 3	not specified	NA

dataset	column	actual_type	n_missing	n_unique	sample_values	expected_class	class_ok
football_data	away_shots_on_target	numeric	[illegible]	[illegible]	2, 5, 7, 3, 9	not specified	NA
football_data	away_yellow_cards	numeric	[illegible]	[illegible]	5, 0, 1, 2, 3	not specified	NA
football_data	full_time_result	character	[illegible]	[illegible]	H, D, A	not specified	NA
football_data	half_time_away_score	numeric	[illegible]	[illegible]	0, 2, 1, 3, 4	not specified	NA
football_data	half_time_home_score	numeric	[illegible]	[illegible]	0, 1, 2, 3, 4	not specified	NA
football_data	half_time_result	character	[illegible]	[illegible]	D, H, A	not specified	NA
football_data	home_corners	numeric	[illegible]	[illegible]	7, 4, 5, 6, 3	not specified	NA
football_data	home_fouls	numeric	[illegible]	[illegible]	25, 13, 22, 0, 9	not specified	NA
football_data	home_red_cards	numeric	[illegible]	[illegible]	0, 1, 2, 3	not specified	NA
football_data	home_shots	numeric	[illegible]	[illegible]	17, 14, 15, 18, 11	not specified	NA
football_data	home_shots_on_target	numeric	[illegible]	[illegible]	7, 6, 5, 2, 3	not specified	NA
football_data	home_yellow_cards	numeric	[illegible]	[illegible]	1, 2, 0, 3, 4	not specified	NA
football_data	league_code	character	[illegible]	[illegible]	D1, D2, E0, E1, E2	not specified	NA

dataset	column	actual_class	typeof	n_missing	n_unique	sample_values	expected_class	class_ok
football_data_seasons	season_code	character	character	0	31	0001, 0102, 0203, 0304, 0405	not specified	NA
international_results	away_score	integer	integer	0	22	0, 2, 1, 3, 4	integer	TRUE
international_results	away_team	character	character	0	321	England, Scotland, Wales, Northern Ireland, Canada	character	TRUE
international_results	away_win	integer	integer	0	2	0, 1	integer	TRUE
international_results	competition	character	character	0	198	Friendly, British Home Championship, Évence Coppée Trophy, Muratti Vase, Copa Lipton	character	TRUE
international_results	date	Date	double	0	16434	1872-11-30, 1873-03-08, 1874-03-07, 1875-03-06, 1876-03-04	Date	TRUE
international_results	draw	integer	integer	0	2	1, 0	integer	TRUE
international_results	goal_difference	integer	integer	0	47	0, 2, 1, 3, 4	integer	TRUE
international_results	home_score	integer	integer	0	26	0, 4, 2, 3, 1	integer	TRUE
international_results	home_team	character	character	0	327	Scotland, England, Wales, Northern Ireland, United States	character	TRUE
international_results	home_win	integer	integer	0	2	0, 1	integer	TRUE
international_results	match_result	character	character	0	3	D, H, A	character	TRUE
international_results	neutral	logical	logical	0	2	FALSE, TRUE	logical	TRUE
international_results	raw_file	character	character	0	1	/Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/international_results/results.csv	character	TRUE
international_results	result_index	integer	integer	0	3	0, 1, -1	integer	TRUE
international_results	season	character	character	0	155	1872, 1873, 1874, 1875, 1876	character	TRUE
international_results	source	character	character	0	1	international_results	character	TRUE
international_results	source_match_id	character	character	0	49257	1872_11_30_scotland_england_friendly, 1873_03_08_england_scotland_friendly, 1874_03_07_scotland_england_friendly, 1875_03_06_england_scotland_friendly, 1876_03_04_scotland_england_friendly	character	TRUE
international_results	total_goals	integer	integer	0	26	0, 6, 3, 4, 2	integer	TRUE
international_results	city	character	character	0	2138	Glasgow, London, Wrexham, Blackburn, Belfast	not specified	NA
international_results	country	character	character	0	269	Scotland, England, Wales, Ireland, United States	not specified	NA
international_results	tournament	character	character	0	198	Friendly, British Home Championship, Évence Coppée Trophy, Muratti Vase, Copa Lipton	not specified	NA
statsbomb_competitions	competition_gender	character	character	0	2	male, female	character	TRUE
statsbomb_competitions	competition_id	integer	integer	0	24	9, 1267, 16, 223, 87	integer	TRUE
statsbomb_competitions	competition_international	logical	logical	0	2	FALSE, TRUE	logical	TRUE
statsbomb_competitions	competition_name	character	character	0	24	1. Bundesliga, African Cup of Nations, Champions League, Copa America, Copa del Rey	character	TRUE
statsbomb_competitions	competition_youth	logical	logical	0	2	FALSE, TRUE	logical	TRUE
statsbomb_competitions	country_name	character	character	0	13	Germany, Africa, Europe, South America, Spain	character	TRUE
statsbomb_competitions	match_available	character	character	0	80	2024-09-28T20:46:38.893391, 2024-05-19T11:11:14.192381, 2026-05-12T21:18:08.827431, 2026-05-15T15:54:04.598614, 2024-02-13T02:35:28.134882	character	TRUE

dataset	column	actual_class	actual_type	n_missing	n_unique	example_values	expected_class	class_ok
statsbomb_competitions	match_available_360	character	character	1683	363	2025-11-15T23:17:41.827093, 2026-05-02T02:07:18.902396, 2026-05-04T01:53:40.309717, 2025-11-02T03:55:53.923193, 2026-04-06T18:59:05.942748	character	TRUE
statsbomb_competitions	match_updated	character	character	0	80	2024-09-28T20:46:38.893391, 2024-05-19T11:11:14.192381, 2026-05-12T21:18:08.827431, 2026-05-15T15:54:04.598614, 2024-02-13T02:35:28.134882	character	TRUE
statsbomb_competitions	match_updated_360	character	character	2236	16	2025-11-15T23:17:41.827093, 2026-05-02T02:07:18.902396, 2021-06-13T16:17:31.694, 2026-05-04T01:53:40.309717, 2021-06-12T16:17:31.694	character	TRUE
statsbomb_competitions	season_id	integer	integer	0	48	281, 27, 107, 4, 1	integer	TRUE
statsbomb_competitions	season_name	character	character	0	48	2023/2024, 2015/2016, 2023, 2018/2019, 2017/2018	character	TRUE
statsbomb_matches	away_score	integer	integer	0	10	1, 4, 2, 0, 3	integer	TRUE
statsbomb_matches	away_team	character	character	0	343	Seattle Sounders, Valencia, Barcelona, Real Madrid, Athletic Club	character	TRUE
statsbomb_matches	away_win	integer	integer	0	2	0, 1	integer	TRUE
statsbomb_matches	competition	character	character	0	24	North American League, La Liga, Indian Super league, African Cup of Nations, Serie A	character	TRUE
statsbomb_matches	date	Date	double	0	1362	1977-08-28, 2018-04-14, 2018-04-29, 2018-05-06, 2018-03-18	Date	TRUE
statsbomb_matches	draw	integer	integer	0	2	0, 1	integer	TRUE
statsbomb_matches	goal_difference	integer	integer	0	21	1, -2, 0, 2, 3	integer	TRUE
statsbomb_matches	home_score	integer	integer	0	13	2, 1, 0, 3, 6	integer	TRUE
statsbomb_matches	home_team	character	character	0	344	NY Cosmos, Barcelona, RC Deportivo La Coruña, Las Palmas, Eibar	character	TRUE
statsbomb_matches	home_win	integer	integer	0	2	1, 0	integer	TRUE
statsbomb_matches	match_result	character	character	0	3	H, A, D	character	TRUE
statsbomb_matches	neutral	logical	logical	3961	1		logical	TRUE
statsbomb_matches	raw_file	character	character	0	80	/Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/116_68.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_1.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_2.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_21.json, /Users/marco/Documents/Projects/worldcup-forecast-r/data/raw/statsbomb_open/matches/11_22.json	character	TRUE
statsbomb_matches	result_num	integer	integer	0	3	1, -1, 0	integer	TRUE
statsbomb_matches	season	character	character	0	48	1977, 2017/2018, 2016/2017, 2009/2010, 2010/2011	character	TRUE
statsbomb_matches	source	character	character	0	1	StatsBomb Open Data	character	TRUE
statsbomb_matches	source_match_id	character	character	0	3961	3750234, 9880, 9912, 9924, 9855	character	TRUE
statsbomb_matches	total_goals	integer	integer	0	14	3, 6, 4, 2, 7	integer	TRUE
statsbomb_matches	competition_country	character	character	0	13	North and Central America, Spain, India, Africa, Italy	not specified	NA
statsbomb_matches	competition_id	integer	integer	0	24	116, 11, 1238, 1267, 12	not specified	NA
statsbomb_matches	data_version	character	character	2	4	1.1.0, 1.0.3, 1.0.2	not specified	NA

dataset	column	actual_class	typeof	n_missing	n_unique	example_values	expected_class	class_ok
statsbomb_matches	match_week	integer	integer	0	38	1, 32, 35, 36, 29	not specified	NA
statsbomb_matches	season_id	integer	integer	0	48	68, 1, 2, 21, 22	not specified	NA
statsbomb_matches	shot_fidelity_version	character	character	1057	2	2	not specified	NA
statsbomb_matches	stadium_name	character	character	10	328	Providence Park, Spotify Camp Nou, Estadio Abanca-Riazor, Estadio de Gran Canaria, Estadio Municipal de Ipurúa	not specified	NA
statsbomb_matches	xy_fidelity_version	character	character	2448	2	2	not specified	NA

```r
failed_class_checks <- class_validation |>
  dplyr::filter(class_ok == FALSE)

knitr::kable(
  failed_class_checks,
  caption = "Columns with failed class checks"
)
```

Table 5: Columns with failed class checks

dataset	column	actual_class	typeof	n_missing	n_unique	example_values	expected_class	class_ok

6. Compact examples from each dataset

These examples are intentionally narrowed to the most important fields so the PDF stays readable.

6.1 StatsBomb competitions

```r
processed_data$statsbomb_competitions |>
  dplyr::slice_head(n = 10) |>
  knitr::kable(
    caption = "First 10 rows: StatsBomb competitions"
  )
```

Table 6: First 10 rows: StatsBomb competitions

competition_id	season_id	country_name	competition_name	competition_gender	competition_youth	competition_international	season_name	match_updated	match_updated_360	match_available_360	match_available
9	281	Germany	1. Bundesliga	male	FALSE	FALSE	2023/2024	2024-09-28T20:46:38.893391	2025-11-15T23:17:41.872093	2025-11-15T23:17:41.872093	2024-09-28T20:46:38.893391
9	27	Germany	1. Bundesliga	male	FALSE	FALSE	2015/2016	2024-05-19T11:11:14.192381	NA	NA	2024-05-19T11:11:14.192381
1267	107	Africa	African Cup of Nations	male	FALSE	TRUE	2023	2026-05-12T21:18:08.827431	2026-05-02T23:07:18.902396	2026-05-02T23:07:18.902396	2026-05-12T21:18:08.827431
16	4	Europe	Champions League	male	FALSE	FALSE	2018/2019	2026-05-15T15:54:04.598614	2021-06-13T16:17:31.694	NA	2026-05-15T15:54:04.598614
16	1	Europe	Champions League	male	FALSE	FALSE	2017/2018	2024-02-13T02:35:28.134882	2021-06-13T16:17:31.694	NA	2024-02-13T02:35:28.134882
16	2	Europe	Champions League	male	FALSE	FALSE	2016/2017	2024-02-13T02:37:32.205154	2021-06-13T16:17:31.694	NA	2024-02-13T02:37:32.205154
16	27	Europe	Champions League	male	FALSE	FALSE	2015/2016	2024-06-12T07:45:38.786894	2021-06-13T16:17:31.694	NA	2024-06-12T07:45:38.786894
16	26	Europe	Champions League	male	FALSE	FALSE	2014/2015	2024-02-12T12:49:54.914228	2021-06-13T16:17:31.694	NA	2024-02-12T12:49:54.914228
16	25	Europe	Champions League	male	FALSE	FALSE	2013/2014	2024-02-12T12:48:48.479157	2021-06-13T16:17:31.694	NA	2024-02-12T12:48:48.479157
16	24	Europe	Champions League	male	FALSE	FALSE	2012/2013	2024-02-12T12:47:34.340413	2021-06-13T16:17:31.694	NA	2024-02-12T12:47:34.340413

6.2 StatsBomb matches

```
processed_data$statsbomb_matches |>
  dplyr::select(
    source_match_id,
    date,
    season,
    competition,
    home_team,
    away_team,
    home_score,
    away_score,
    match_result,
    result_class,
    neutral
  ) |>
  dplyr::slice_head(n = 10) |>
  knitr::kable(
    caption = "First 10 rows: StatsBomb matches"
  )
```

Table 7: First 10 rows: StatsBomb matches

source_match_id	date	season	competition	home_team	away_team	home_score	away_score	match_result	result_class	neutral
3750234	1977-08-28	1977	North American League	NY Cosmos	Seattle Sounders	2	1	H	1	NA
9880	2018-04-14	2017/2018	La Liga	Barcelona	Valencia	2	1	H	1	NA
9912	2018-04-29	2017/2018	La Liga	RC Deportivo La Coruña	Barcelona	2	4	A	-1	NA
9924	2018-05-06	2017/2018	La Liga	Barcelona	Real Madrid	2	2	D	0	NA
9855	2018-03-18	2017/2018	La Liga	Barcelona	Athletic Club	2	0	H	1	NA
9827	2018-03-01	2017/2018	La Liga	Las Palmas	Barcelona	1	1	D	0	NA
9799	2018-02-17	2017/2018	La Liga	Eibar	Barcelona	0	2	A	-1	NA
9636	2017-10-01	2017/2018	La Liga	Barcelona	Las Palmas	3	0	H	1	NA
9609	2017-09-19	2017/2018	La Liga	Barcelona	Eibar	6	1	H	1	NA
9575	2017-08-20	2017/2018	La Liga	Barcelona	Real Betis	2	0	H	1	NA

6.3 football-data.co.uk matches

```r
processed_data$football_data_uk_matches |>
  dplyr::select(
    source_match_id,
    date,
    season,
    competition,
    home_team,
    away_team,
    home_score,
    away_score,
    match_result,
    result_class,
    full_time_result,
    home_shots,
    away_shots,
    home_shots_on_target,
    away_shots_on_target
  ) |>
  dplyr::slice_head(n = 10) |>
  knitr::kable(
    caption = "First 10 rows: football-data.co.uk matches"
  )
```

Table 8: First 10 rows: football-data.co.uk matches

source_match_id	date	season	competition	home_team	away_team	home_score	away_score	match_result	result_class	full_time_result	home_shots	away_shots	home_shots_on_target	away_shots_on_target
D1_0001_001	2000-08-11	0001	D1	Dortmund	Hansa Rostock	1	0	H	1	H	17	5	7	2
D1_0001_002	2000-08-12	0001	D1	Bayern Munich	Hertha	4	1	H	1	H	14	11	6	5
D1_0001_003	2000-08-12	0001	D1	Freiburg	Stuttgart	4	0	H	1	H	15	18	7	5
D1_0001_004	2000-08-12	0001	D1	Hamburg	Munich 1860	2	2	D	0	D	18	9	5	7
D1_0001_005	2000-08-12	0001	D1	Kaiserslautern	Bochum	0	1	A	-1	A	11	5	2	2
D1_0001_006	2000-08-12	0001	D1	Leverkusen	Wolfsburg	2	0	H	1	H	18	5	5	5
D1_0001_007	2000-08-12	0001	D1	Werder Bremen	Cottbus	3	1	H	1	H	12	3	2	3
D1_0001_008	2000-08-13	0001	D1	Ein Frankfurt	Unterhaching	3	0	H	1	H	7	15	5	3
D1_0001_009	2000-08-13	0001	D1	Schalke 04	FC Koln	2	1	H	1	H	16	10	6	3
D1_0001_010	2000-08-18	0001	D1	Cottbus	Dortmund	1	4	A	-1	A	11	10	3	5

6.4 International results

```
processed_data$international_results |>
  dplyr::select(
    source_match_id,
    date,
    season,
    tournament,
    home_team,
    away_team,
    home_score,
    away_score,
    match_result,
    result_class,
    neutral,
    city,
    country
```

```r
  ) |>
  dplyr::slice_head(n = 10) |>
  knitr::kable(
    caption = "First 10 rows: international results"
  )
```

Table 9: First 10 rows: international results

source_match_id	date	season	tournament	home_team	away_team	home_score	away_score	match_result	result_class	neutral	city	country
1872_11_30_scotland_england_friendly	1872-11-30	1872	Friendly	Scotland	England	0	0	D	0	FALSE	Glasgow	Scotland
1873_03_08_england_scotland_friendly	1873-03-08	1873	Friendly	England	Scotland	4	2	H	1	FALSE	London	England
1874_03_07_scotland_england_friendly	1874-03-07	1874	Friendly	Scotland	England	2	1	H	1	FALSE	Glasgow	Scotland
1875_03_06_england_scotland_friendly	1875-03-06	1875	Friendly	England	Scotland	2	2	D	0	FALSE	London	England
1876_03_04_scotland_england_friendly	1876-03-04	1876	Friendly	Scotland	England	3	0	H	1	FALSE	Glasgow	Scotland
1876_03_25_scotland_wales_friendly	1876-03-25	1876	Friendly	Scotland	Wales	4	0	H	1	FALSE	Glasgow	Scotland
1877_03_03_england_scotland_friendly	1877-03-03	1877	Friendly	England	Scotland	1	3	A	-1	FALSE	London	England
1877_03_05_wales_scotland_friendly	1877-03-05	1877	Friendly	Wales	Scotland	0	2	A	-1	FALSE	Wrexham	Wales
1878_03_02_scotland_england_friendly	1878-03-02	1878	Friendly	Scotland	England	7	2	H	1	FALSE	Glasgow	Scotland
1878_03_23_scotland_wales_friendly	1878-03-23	1878	Friendly	Scotland	Wales	9	0	H	1	FALSE	Glasgow	Scotland

7. Random examples from each match dataset

First rows can be misleading because they often come from one competition or one time period. Random examples help verify broader plausibility.

```r
set.seed(2026)

match_tables <- processed_data[c(
  "statsbomb_matches",
  "football_data_uk_matches",
  "international_results"
```

```r
)]

random_match_examples <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    dat |>
      dplyr::select(
        source_match_id,
        date,
        competition,
        home_team,
        away_team,
        home_score,
        away_score,
        match_result,
        result_class,
        home_win,
        draw,
        away_win,
        goal_difference,
        total_goals,
        neutral
      ) |>
      dplyr::slice_sample(n = min(10, nrow(dat))) |>
      dplyr::mutate(dataset = dataset_name, .before = 1)
  }
)

knitr::kable(
  random_match_examples,
  caption = "Random examples from match datasets"
)
```

Table 10: Random examples from match datasets

dataset	source_match_id	date	competition	home_team	away_team	home_score	away_score	match_result	result_class	home_win	draw	away_win	goal_difference	total_goals	neutral
statsbomb_matches	16205	2019-03-09	La Liga	Barcelona	Rayo Vallecano	3	1	H	1	1	0	0	2	4	NA
statsbomb_matches	3900530	2015-12-01	Ligue 1	Nantes	Lyon	0	0	D	0	0	1	0	0	0	NA
statsbomb_matches	3920415	2024-01-23	African Cup of Nations	Mauritania	Algeria	1	0	H	1	1	0	0	1	1	NA
statsbomb_matches	3754349	2016-03-02	Premier League	West Ham United	Tottenham Hotspur	1	0	H	1	1	0	0	1	1	NA
statsbomb_matches	3901176	2016-03-12	Ligue 1	Toulouse	Bordeaux	4	0	H	1	1	0	0	4	4	NA

dataset	source_match_id	date	competition	home_team	away_team	home_score	away_score	result	[illegible]	home_win	draw	away_win	goal_difference	total_goals	neutral
statsbomb_matches	3912202	2024-02-11	Frauen Bundesliga	SC Freiburg	Duisburg WFC	1	1	D	0	0	1	0	0	2	NA
statsbomb_matches	3910994	2023-11-26	Liga F	Athletic Club Bilbao	Barcelona WFC	0	4	A	-1	0	0	1	-4	4	NA
statsbomb_matches	3835327	2022-07-11	UEFA Women's Euro	England Women's	Norway Women's	8	0	H	1	1	0	0	8	8	NA
statsbomb_matches	69327	2012-01-15	La Liga	Barcelona	Real Betis	4	2	H	1	1	0	0	2	6	NA
statsbomb_matches	3879798	2016-04-02	Serie A	Juventus	Empoli	1	0	H	1	1	0	0	1	1	NA
football_data_uk_matches	N1_0708_235	2008-02-29	N1	Vitesse	VVV Venlo	1	3	A	-1	0	0	1	-2	4	NA
football_data_uk_matches	E1_1516_76	2015-09-15	E1	Cardiff	Hull	0	2	A	-1	0	0	1	-2	2	NA
football_data_uk_matches	F1_2223_81	2022-09-30	F1	Angers	Marseille	0	3	A	-1	0	0	1	-3	3	NA
football_data_uk_matches	E1_0001_142	2000-10-18	E1	Wimbledon	Blackburn	0	2	A	-1	0	0	1	-2	2	NA
football_data_uk_matches	SP1_0304_222	2004-02-07	SP1	Valencia	Ath Madrid	3	0	H	1	1	0	0	3	3	NA
football_data_uk_matches	SP1_1112_212	2012-02-11	SP1	Osasuna	Barcelona	3	2	H	1	1	0	0	1	5	NA
football_data_uk_matches	E2_1314_96	2013-09-23	E2	Brentford	Leyton Orient	0	2	A	-1	0	0	1	-2	2	NA
football_data_uk_matches	N1_1213_279	2013-04-21	N1	VVV Venlo	Twente	2	2	D	0	0	1	0	0	4	NA
football_data_uk_matches	N1_0607_194	2007-01-27	N1	Willem II	Vitesse	0	0	D	0	0	1	0	0	0	NA
football_data_uk_matches	E2_9798_551	1998-05-02	E2	Walsall	Wycombe	0	1	A	-1	0	0	1	-1	1	NA
international_results	2005_06_18_kenya_morocco_fifa_world_cup_qualification	2005-06-18	FIFA World Cup qualification	Kenya	Morocco	0	0	D	0	0	1	0	0	0	FALSE

dataset	source_match_id	date	competition	home_team	away_team	home_score	away_score	match_result	result_class	home_win	draw	away_win	goal_difference	total_goals	neutral
international_results	2004_12_02_thailand_slovakia_king_s_cup	2004-12-02	King's Cup	Thailand	Slovakia	1	1	D	0	0	1	0	0	2	FALSE
international_results	1938_01_11_dominica_saint_lucia_friendly	1938-01-11	Friendly	Dominica	Saint Lucia	1	0	H	1	1	0	0	1	1	FALSE
international_results	1993_08_15_congo_guinea_african_cup_of_nations_qualification	1993-08-15	African Cup of Nations qualification	Congo	Guinea	0	0	D	0	0	1	0	0	0	FALSE
international_results	1986_10_05_south_korea_saudi_arabia_asian_games	1986-10-05	Asian Games	South Korea	Saudi Arabia	2	0	H	1	1	0	0	2	2	FALSE
international_results	2018_04_08_barawa_chagos_islands_friendly	2018-04-08	Friendly	Barawa	Chagos Islands	4	1	H	1	1	0	0	3	5	FALSE
international_results	2024_01_16_saudi_arabia_oman_afc_asian_cup	2024-01-16	AFC Asian Cup	Saudi Arabia	Oman	2	1	H	1	1	0	0	1	3	TRUE
international_results	1947_10_18_wales_england_british_home_championship	1947-10-18	British Home Championship	Wales	England	0	3	A	-1	0	0	1	-3	3	FALSE
international_results	1982_03_10_ethiopia_zambia_african_cup_of_nations	1982-03-10	African Cup of Nations	Ethiopia	Zambia	0	1	A	-1	0	0	1	-1	1	TRUE
international_results	2006_03_01_south_korea_angola_friendly	2006-03-01	Friendly	South Korea	Angola	1	0	H	1	1	0	0	1	1	FALSE

8. Outcome logic validation

This verifies that the derived outcome columns agree with the raw scores.

Expected logic:

- home score greater than away score means `match_result == "H"` and `result_class == 1`
- home score equal to away score means `match_result == "D"` and `result_class == 0`
- home score less than away score means `match_result == "A"` and `result_class == -1`
- `goal_difference` should equal `home_score - away_score`
- `total_goals` should equal `home_score + away_score`

```r
outcome_logic_check <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    dat |>
      dplyr::mutate(
        expected_match_result = dplyr::case_when(
          home_score > away_score ~ "H",
```

```
          home_score == away_score ~ "D",
          home_score < away_score ~ "A",
          TRUE ~ NA_character_
        ),
        expected_result_class = dplyr::case_when(
          expected_match_result == "H" ~ 1L,
          expected_match_result == "D" ~ 0L,
          expected_match_result == "A" ~ -1L,
          TRUE ~ NA_integer_
        ),
        expected_home_win = as.integer(expected_match_result == "H"),
        expected_draw = as.integer(expected_match_result == "D"),
        expected_away_win = as.integer(expected_match_result == "A"),
        expected_goal_difference = home_score - away_score,
        expected_total_goals = home_score + away_score,
        result_logic_ok =
          match_result == expected_match_result &
          result_class == expected_result_class &
          home_win == expected_home_win &
          draw == expected_draw &
          away_win == expected_away_win &
          goal_difference == expected_goal_difference &
          total_goals == expected_total_goals
      ) |>
      dplyr::summarise(
        dataset = dataset_name,
        rows = dplyr::n(),
        failed_rows = sum(!result_logic_ok, na.rm = TRUE),
        missing_logic_rows = sum(is.na(result_logic_ok))
      )
  }
)

knitr::kable(
  outcome_logic_check,
  caption = "Outcome logic validation summary"
)
```

Table 11: Outcome logic validation summary

dataset	rows	failed_rows	missing_logic_rows
statsbomb_matches	3961	0	0
football_data_uk_matches	177295	0	0
international_results	49257	0	0

```
failed_outcome_examples <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    dat |>
      dplyr::mutate(
        expected_match_result = dplyr::case_when(
          home_score > away_score ~ "H",
```

```
        home_score == away_score ~ "D",
        home_score < away_score ~ "A",
        TRUE ~ NA_character_
      ),
      expected_result_class = dplyr::case_when(
        expected_match_result == "H" ~ 1L,
        expected_match_result == "D" ~ 0L,
        expected_match_result == "A" ~ -1L,
        TRUE ~ NA_integer_
      ),
      expected_home_win = as.integer(expected_match_result == "H"),
      expected_draw = as.integer(expected_match_result == "D"),
      expected_away_win = as.integer(expected_match_result == "A"),
      expected_goal_difference = home_score - away_score,
      expected_total_goals = home_score + away_score,
      result_logic_ok =
        match_result == expected_match_result &
        result_class == expected_result_class &
        home_win == expected_home_win &
        draw == expected_draw &
        away_win == expected_away_win &
        goal_difference == expected_goal_difference &
        total_goals == expected_total_goals
    ) |>
    dplyr::filter(!result_logic_ok | is.na(result_logic_ok)) |>
    dplyr::select(
      source_match_id,
      date,
      competition,
      home_team,
      away_team,
      home_score,
      away_score,
      match_result,
      expected_match_result,
      result_class,
      expected_result_class,
      home_win,
      expected_home_win,
      draw,
      expected_draw,
      away_win,
      expected_away_win,
      goal_difference,
      expected_goal_difference,
      total_goals,
      expected_total_goals
    ) |>
    dplyr::slice_head(n = 20) |>
    dplyr::mutate(dataset = dataset_name, .before = 1)
  }
)
```

```r
knitr::kable(
  failed_outcome_examples,
  caption = "Failed outcome examples, if any"
)
```

Table 12: Failed outcome examples, if any

dataset	[illegible]	total_goa

9. Missingness summary

```r
missingness_summary <- purrr::imap_dfr(
  processed_data,
  function(dat, dataset_name) {
    tibble::tibble(
      dataset = dataset_name,
      column = names(dat),
      n_rows = nrow(dat),
      n_missing = purrr::map_int(dat, ~ sum(is.na(.x))),
      pct_missing = round(100 * n_missing / n_rows, 2)
    )
  }
) |>
  dplyr::arrange(dataset, dplyr::desc(pct_missing), column)

knitr::kable(
  missingness_summary,
  caption = "Missingness by dataset and column"
)
```

Table 13: Missingness by dataset and column

dataset	column	n_rows	n_missing	pct_missing
football_data_uk_matches	neutral	177295	177295	100.00
football_data_uk_matches	away_fouls	177295	80336	45.31
football_data_uk_matches	home_fouls	177295	80336	45.31
football_data_uk_matches	away_shots_on_target	177295	80114	45.19
football_data_uk_matches	home_shots_on_target	177295	80117	45.19
football_data_uk_matches	away_corners	177295	79571	44.88
football_data_uk_matches	home_corners	177295	79571	44.88
football_data_uk_matches	away_shots	177295	79196	44.67
football_data_uk_matches	home_shots	177295	79199	44.67
football_data_uk_matches	away_red_cards	177295	79014	44.57
football_data_uk_matches	away_yellow_cards	177295	79014	44.57
football_data_uk_matches	home_red_cards	177295	79014	44.57
football_data_uk_matches	home_yellow_cards	177295	79015	44.57
football_data_uk_matches	half_time_away_score	177295	20042	11.30
football_data_uk_matches	half_time_home_score	177295	20041	11.30

dataset	column	n_rows	n_missing	pct_missing
football_data_uk_matches	half_time_result	177295	20041	11.30
football_data_uk_matches	away_score	177295	0	0.00
football_data_uk_matches	away_team	177295	0	0.00
football_data_uk_matches	away_win	177295	0	0.00
football_data_uk_matches	competition	177295	0	0.00
football_data_uk_matches	date	177295	0	0.00
football_data_uk_matches	draw	177295	0	0.00
football_data_uk_matches	full_time_result	177295	0	0.00
football_data_uk_matches	goal_difference	177295	0	0.00
football_data_uk_matches	home_score	177295	0	0.00
football_data_uk_matches	home_team	177295	0	0.00
football_data_uk_matches	home_win	177295	0	0.00
football_data_uk_matches	match_result	177295	0	0.00
football_data_uk_matches	raw_file	177295	0	0.00
football_data_uk_matches	result_class	177295	0	0.00
football_data_uk_matches	season	177295	0	0.00
football_data_uk_matches	source	177295	0	0.00
football_data_uk_matches	source_league_code	177295	0	0.00
football_data_uk_matches	source_match_id	177295	0	0.00
football_data_uk_matches	source_season_code	177295	0	0.00
football_data_uk_matches	total_goals	177295	0	0.00
international_results	away_score	49257	0	0.00
international_results	away_team	49257	0	0.00
international_results	away_win	49257	0	0.00
international_results	city	49257	0	0.00
international_results	competition	49257	0	0.00
international_results	country	49257	0	0.00
international_results	date	49257	0	0.00
international_results	draw	49257	0	0.00
international_results	goal_difference	49257	0	0.00
international_results	home_score	49257	0	0.00
international_results	home_team	49257	0	0.00
international_results	home_win	49257	0	0.00
international_results	match_result	49257	0	0.00
international_results	neutral	49257	0	0.00
international_results	raw_file	49257	0	0.00
international_results	result_class	49257	0	0.00
international_results	season	49257	0	0.00
international_results	source	49257	0	0.00
international_results	source_match_id	49257	0	0.00
international_results	total_goals	49257	0	0.00
international_results	tournament	49257	0	0.00
statsbomb_competitions	match_available_360	80	68	85.00
statsbomb_competitions	match_updated_360	80	22	27.50
statsbomb_competitions	competition_gender	80	0	0.00
statsbomb_competitions	competition_id	80	0	0.00
statsbomb_competitions	competition_international	80	0	0.00
statsbomb_competitions	competition_name	80	0	0.00
statsbomb_competitions	competition_youth	80	0	0.00
statsbomb_competitions	country_name	80	0	0.00
statsbomb_competitions	match_available	80	0	0.00
statsbomb_competitions	match_updated	80	0	0.00

dataset	column	n_rows	n_missing	pct_missing
statsbomb_competitions	season_id	80	0	0.00
statsbomb_competitions	season_name	80	0	0.00
statsbomb_matches	neutral	3961	3961	100.00
statsbomb_matches	xy_fidelity_version	3961	243	6.13
statsbomb_matches	shot_fidelity_version	3961	197	4.97
statsbomb_matches	stadium	3961	10	0.25
statsbomb_matches	data_version	3961	2	0.05
statsbomb_matches	away_score	3961	0	0.00
statsbomb_matches	away_team	3961	0	0.00
statsbomb_matches	away_win	3961	0	0.00
statsbomb_matches	competition	3961	0	0.00
statsbomb_matches	competition_country	3961	0	0.00
statsbomb_matches	competition_id	3961	0	0.00
statsbomb_matches	date	3961	0	0.00
statsbomb_matches	draw	3961	0	0.00
statsbomb_matches	goal_difference	3961	0	0.00
statsbomb_matches	home_score	3961	0	0.00
statsbomb_matches	home_team	3961	0	0.00
statsbomb_matches	home_win	3961	0	0.00
statsbomb_matches	match_result	3961	0	0.00
statsbomb_matches	match_week	3961	0	0.00
statsbomb_matches	raw_file	3961	0	0.00
statsbomb_matches	result_class	3961	0	0.00
statsbomb_matches	season	3961	0	0.00
statsbomb_matches	season_id	3961	0	0.00
statsbomb_matches	source	3961	0	0.00
statsbomb_matches	source_match_id	3961	0	0.00
statsbomb_matches	total_goals	3961	0	0.00

10. Duplicate checks

These are basic identity checks. Some repeated fixtures may be legitimate, but they should be inspected.

```
duplicate_checks <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    duplicate_source_ids <- dat |>
      dplyr::count(source_match_id, name = "n") |>
      dplyr::filter(n > 1L)

    duplicate_fixture_keys <- dat |>
      dplyr::count(
        date,
        home_team,
        away_team,
        competition,
        name = "n"
      ) |>
      dplyr::filter(n > 1L)
```

```
    exact_duplicate_rows <- dat |>
      dplyr::count(
        dplyr::across(dplyr::everything()),
        name = "n"
      ) |>
      dplyr::filter(n > 1L)

    tibble::tibble(
      dataset = dataset_name,
      duplicate_source_match_ids = nrow(duplicate_source_ids),
      duplicate_date_home_away_competition_keys = nrow(duplicate_fixture_keys),
      exact_duplicate_rows = nrow(exact_duplicate_rows)
    )
  }
)

knitr::kable(
  duplicate_checks,
  caption = "Duplicate checks for match datasets"
)
```

Table 14: Duplicate checks for match datasets

dataset	duplicate_source_match_ids	duplicate_date_home_away_competition_keys	exact_duplicate_rows
statsbomb_matches	0	0	0
football_data_uk_matches	0	1222	0
international_results	0	1	0

11. football-data.co.uk numeric stat validation

Some football-data.co.uk columns are read as doubles because historical files can be inconsistent. This check verifies whether those columns are still integer-valued.

```
football_data <- processed_data$football_data_uk_matches

integer_valued_stat_cols <- c(
  "half_time_home_score",
  "half_time_away_score",
  "home_shots",
  "away_shots",
  "home_shots_on_target",
  "away_shots_on_target",
  "home_corners",
  "away_corners",
  "home_fouls",
  "away_fouls",
  "home_yellow_cards",
  "away_yellow_cards",
  "home_red_cards",
  "away_red_cards"
```

```r
)

integer_valued_stat_cols <- intersect(integer_valued_stat_cols, names(football_data))

integer_valued_check <- tibble::tibble(
  column = integer_valued_stat_cols,
  class = purrr::map_chr(
    football_data[integer_valued_stat_cols],
    ~ paste(class(.x), collapse = " / ")
  ),
  n_missing = purrr::map_int(
    football_data[integer_valued_stat_cols],
    ~ sum(is.na(.x))
  ),
  non_integer_values = purrr::map_int(
    football_data[integer_valued_stat_cols],
    ~ sum(!is.na(.x) & .x != floor(.x))
  ),
  min_value = purrr::map_dbl(
    football_data[integer_valued_stat_cols],
    ~ suppressWarnings(min(.x, na.rm = TRUE))
  ),
  max_value = purrr::map_dbl(
    football_data[integer_valued_stat_cols],
    ~ suppressWarnings(max(.x, na.rm = TRUE))
  )
)

knitr::kable(
  integer_valued_check,
  caption = "Integer-valued check for football-data.co.uk numeric match stats"
)
```

Table 15: Integer-valued check for football-data.co.uk numeric match stats

column	class	n_missing	non_integer_values	min_value	max_value
half_time_home_score	numeric	20041	0	0	7
half_time_away_score	numeric	20042	0	0	7
home_shots	numeric	79199	0	0	46
away_shots	numeric	79196	0	0	45
home_shots_on_target	numeric	80117	0	0	27
away_shots_on_target	numeric	80114	0	0	23
home_corners	numeric	79571	0	0	26
away_corners	numeric	79571	0	0	23
home_fouls	numeric	80336	0	0	145
away_fouls	numeric	80336	0	0	77
home_yellow_cards	numeric	79015	0	0	11
away_yellow_cards	numeric	79014	0	0	10
home_red_cards	numeric	79014	0	0	3
away_red_cards	numeric	79014	0	0	9

12. Result distribution

This gives a quick sanity check of home wins, draws, and away wins.

```r
result_distribution <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    dat |>
      dplyr::count(match_result, name = "matches") |>
      dplyr::mutate(
        dataset = dataset_name,
        pct = round(100 * matches / sum(matches), 2),
        .before = 1
      )
  }
)

knitr::kable(
  result_distribution,
  caption = "Match result distribution"
)
```

Table 16: Match result distribution

dataset	pct	match_result	matches
statsbomb_matches	33.00	A	1307
statsbomb_matches	22.19	D	879
statsbomb_matches	44.81	H	1775
football_data_uk_matches	27.34	A	48465
football_data_uk_matches	27.45	D	48673
football_data_uk_matches	45.21	H	80157
international_results	28.27	A	13925
international_results	22.74	D	11202
international_results	48.99	H	24130

13. Score and goals summary

```r
score_summary <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    dat |>
      dplyr::summarise(
        dataset = dataset_name,
        rows = dplyr::n(),
        min_home_score = min(home_score, na.rm = TRUE),
        max_home_score = max(home_score, na.rm = TRUE),
        min_away_score = min(away_score, na.rm = TRUE),
        max_away_score = max(away_score, na.rm = TRUE),
        mean_home_score = round(mean(home_score, na.rm = TRUE), 3),
        mean_away_score = round(mean(away_score, na.rm = TRUE), 3),
```

```
        mean_total_goals = round(mean(total_goals, na.rm = TRUE), 3),
        max_total_goals = max(total_goals, na.rm = TRUE)
      )
  }
)

knitr::kable(
  score_summary,
  caption = "Score and total-goals summary"
)
```

Table 17: Score and total-goals summary

dataset	rows	min_home_score	max_home_score	min_away_score	max_away_score	mean_home_score	mean_away_score	mean_total_goals	max_total_goals
statsbomb_matches	3961	0	13	0	9	1.608	1.288	2.896	13
football_data_uk_matches	177295	0	10	0	13	1.487	1.109	2.596	13
international_results	49257	0	31	0	21	1.757	1.183	2.940	31

14. Extreme score examples

Extreme rows are useful for catching parsing problems or unusual but real matches.

```
extreme_score_examples <- purrr::imap_dfr(
  match_tables,
  function(dat, dataset_name) {
    dat |>
      dplyr::arrange(dplyr::desc(total_goals)) |>
      dplyr::select(
        source_match_id,
        date,
        competition,
        home_team,
        away_team,
        home_score,
        away_score,
        match_result,
        result_class,
        total_goals,
        goal_difference,
        neutral
      ) |>
      dplyr::slice_head(n = 10) |>
      dplyr::mutate(dataset = dataset_name, .before = 1)
  }
)

knitr::kable(
  extreme_score_examples,
  caption = "Highest-total-goal examples by dataset"
)
```

Table 18: Highest-total-goal examples by dataset

dataset	source_match_id	date	competition	home_team	away_team	home_score	away_score	result	home_result	total_goals	goal_difference	neutral
statsbomb_matches	22943	2019-06-11	Women's World Cup	United States Women's	Thailand W	13	0	H	1	13	13	NA
statsbomb_matches	3825700	2015-12-20	La Liga	Real Madrid	Rayo Vallecano	10	2	H	1	12	8	NA
statsbomb_matches	2275104	2019-12-01	FA Women's Super League	Arsenal WFC	Bristol City WFC	11	1	H	1	12	10	NA
statsbomb_matches	3929008	2024-05-01	Serie A Women	Sassuolo W	AS Roma W	5	6	A	-1	11	-1	NA
statsbomb_matches	15973	2018-09-02	La Liga	Barcelona	Huesca	8	2	H	1	10	6	NA
statsbomb_matches	3813317	2021-11-30	Indian Super league	Odisha	East Bengal	6	4	H	1	10	2	NA
statsbomb_matches	3912207	2024-02-17	Frauen Bundesliga	Nürnberg W	VfL Wolfsburg WFC	1	9	A	-1	10	-8	NA
statsbomb_matches	3911293	2024-01-06	Liga F	Barcelona WFC	FC Levante Badalona	9	1	H	1	10	8	NA
statsbomb_matches	3913146	2024-03-03	FA Women's Super League	Bristol City WFC	Brighton & Hove Albion WFC	3	7	A	-1	10	-4	NA
statsbomb_matches	3764238	2020-09-12	FA Women's Super League	West Ham United LFC	Arsenal WFC	1	9	A	-1	10	-8	NA
football_data_uk_matches	N1_2021_45	2020-10-24	N1	VVV Venlo	Ajax	0	13	A	-1	13	-13	NA
football_data_uk_matches	D2_9697_302	1997-06-11	D2	Kaiserslautern	Meppen	7	6	H	1	13	1	NA
football_data_uk_matches	SC0_0910_222	2010-05-05	SC0	Motherwell	Hibernian	6	6	D	0	12	0	NA
football_data_uk_matches	SP2_1415_190	2014-12-20	SP2	Numancia	Lugo	6	6	D	0	12	0	NA
football_data_uk_matches	SP1_1516_158	2015-12-20	SP1	Real Madrid	Vallecano	10	2	H	1	12	8	NA

dataset	source_match_id	date	competition	home_team	away_team	home_score	away_score	match_result	result	total_goals	goal_difference	neutral
football_data_uk_matches	E1_0203_177	2002-10-29	E1	Grimsby	Burnley	6	5	H	1	11	1	NA
football_data_uk_matches	E1_0203_472	2003-04-05	E1	Burnley	Watford	4	7	A	-1	11	-3	NA
football_data_uk_matches	D1_0506_185	2006-02-11	D1	Schalke 04	Leverkusen	7	4	H	1	11	3	NA
football_data_uk_matches	E0_0708_72	2007-09-29	E0	Portsmouth	Reading	7	4	H	1	11	3	NA
football_data_uk_matches	E3_0910_413	2010-03-13	E3	Burton	Cheltenham	5	6	A	-1	11	-1	NA
international_results	2001_04_11_australia_american_samoa_fifa_world_cup_qualification	2001-04-11	FIFA World Cup qualification	Australia	American Samoa	31	0	H	1	31	31	FALSE
international_results	1971_09_13_tahiti_cook_islands_south_pacific_games	1971-09-13	South Pacific Games	Tahiti	Cook Islands	30	0	H	1	30	30	FALSE
international_results	1979_08_30_fiji_kiribati_south_pacific_games	1979-08-30	South Pacific Games	Fiji	Kiribati	24	0	H	1	24	24	FALSE
international_results	2001_04_09_australia_tonga_fifa_world_cup_qualification	2001-04-09	FIFA World Cup qualification	Australia	Tonga	22	0	H	1	22	22	FALSE
international_results	2006_11_24_s_pmi_monaco_viva_world_cup	2006-11-24	Viva World Cup	Sápmi	Monaco	21	1	H	1	22	20	TRUE
international_results	2013_06_24_provence_tibet_international_tournament_of_peoples_cultures_and_tribes	2013-06-24	International Tournament of Peoples, Cultures and Tribes	Provence	Tibet	22	0	H	1	22	22	FALSE
international_results	1966_04_03_libya_oman_arab_cup	1966-04-03	Arab Cup	Libya	Oman	21	0	H	1	21	21	TRUE
international_results	1997_05_13_kazakhstan_guam_east_asian_games	1997-05-13	East Asian Games	Kazakhstan	Guam	20	1	H	1	21	19	TRUE
international_results	2005_03_11_guam_north_korea_eaff_championship	2005-03-11	EAFF Championship	Guam	North Korea	0	21	A	-1	21	-21	TRUE
international_results	2013_06_23_quebec_tibet_international_tournament_of_peoples_cultures_and_tribes	2013-06-23	International Tournament of Peoples, Cultures and Tribes	Quebec	Tibet	21	0	H	1	21	21	TRUE

15. Column display audit

This section shows which columns were omitted from compact example tables. They are still loaded in `processed_data`; they were only omitted from some tables to keep the PDF readable.

```r
displayed_columns <- list(
  statsbomb_competitions = c(
    "competition_id",
    "season_id",
    "country_name",
    "competition_name",
    "competition_gender",
    "competition_youth",
    "competition_international",
    "season_name"
  ),

  statsbomb_matches = c(
    "source_match_id",
    "date",
    "season",
    "competition",
    "home_team",
    "away_team",
    "home_score",
    "away_score",
    "match_result",
    "result_class",
    "neutral"
  ),

  football_data_uk_matches = c(
    "source_match_id",
    "date",
    "season",
    "competition",
    "home_team",
    "away_team",
    "home_score",
    "away_score",
    "match_result",
    "result_class",
    "full_time_result",
    "home_shots",
    "away_shots",
    "home_shots_on_target",
    "away_shots_on_target"
  ),

  international_results = c(
    "source_match_id",
    "date",
    "season",
    "tournament",
```

```r
    "home_team",
    "away_team",
    "home_score",
    "away_score",
    "match_result",
    "result_class",
    "neutral",
    "city",
    "country"
  )
)

omitted_columns_audit <- purrr::imap_dfr(
  processed_data,
  function(dat, dataset_name) {
    shown <- displayed_columns[[dataset_name]]
    all_cols <- names(dat)

    tibble::tibble(
      dataset = dataset_name,
      n_total_columns = length(all_cols),
      n_displayed_columns = length(intersect(shown, all_cols)),
      n_omitted_columns = length(setdiff(all_cols, shown)),
      omitted_columns = paste(setdiff(all_cols, shown), collapse = ", ")
    )
  }
)

knitr::kable(
  omitted_columns_audit,
  caption = "Columns omitted from compact display tables"
)
```

Table 19: Columns omitted from compact display tables

dataset	n_total_columns	n_displayed_columns	n_omitted_columns	omitted_columns
statsbomb_competitions	12	8	4	match_updated, match_updated_360, match_available_360, match_available
statsbomb_matches	26	11	15	source, raw_file, competition_id, season_id, match_week, competition_country, stadium, data_version, shot_fidelity_version, xy_fidelity_version, home_win, draw, away_win, goal_difference, total_goals
football_data_uk_matches	36	15	21	source, raw_file, source_league_code, source_season_code, half_time_home_score, half_time_away_score, half_time_result, home_corners, away_corners, home_fouls, away_fouls, home_yellow_cards, away_yellow_cards, home_red_cards, away_red_cards, home_win, draw, away_win, goal_difference, total_goals, neutral
international_results	21	13	8	source, raw_file, competition, home_win, draw, away_win, goal_difference, total_goals

16. Final interpretation checklist

After knitting this notebook, check the following:

1. All expected processed files exist.
2. All datasets load successfully.
3. Dates are parsed as `Date`.
4. Team names, competition names, tournament names, city names, and country names are character fields.
5. Scores and result flags are integer fields where expected.
6. `neutral` is logical.
7. `match_result`, `result_class`, `home_win`, `draw`, and `away_win` agree with the scores.
8. `goal_difference` equals `home_score - away_score`.
9. `total_goals` equals `home_score + away_score`.
10. Missingness is explainable.
11. Duplicate checks do not reveal unexpected duplicate match identities.
12. Random and extreme examples look like real soccer match records.